

Save the Stream!

Target grade: Kindergarten - Grade 2

Time: 35-45 minutes

Team size: 3-4 students

Introduction

Students investigate how pollution left on the ground can be carried into streams and what type of pollution through a simulated “stream rescue” activity.

Each team will receive a shallow tray containing a model stream with toy fish, plants, and rocks. Streams will also contain materials such as paper scraps, bottle caps, soil, leaves and small amounts of cooking oil (optional). Students are tasked with determining which materials are pollution and waste then removing them from the model streams using tongs.

Learning Objectives

Students will:

- Identify common materials that may pollute
- Explain that pollution on streets, lawns, and sidewalks can become water pollution
- Recognize several different types of water pollution
- Learn how to carefully remove pollution from streams and sort pollution into simple categories

Materials (Per team):

<u>Material</u>	<u>Quantity</u>
Shallow tray, plastic container, or aluminum pan	1
Blue paper, blue plastic, or water	Enough to create 1 stream
Toy fish or laminated fish pictures	3-5
Small plants, twigs, leaves, and small rocks	As needed
Tongs or large tweezers	1-2
Small spoon	1
Sorting cups or bowls	4
Student observation sheet	1 per student or team
Pencils	As needed

Simulated Pollution

<u>Material</u>	<u>Represents</u>
Paper scraps	Paper litter
Plastic bottle caps	Plastic waste
Pieces of yarn or string	Fishing line or discarded string
Small foil pieces	Metal or food-wrapper litter
Soil or cocoa powder	Sediment

Additional materials may always be added to represent additional types of pollution; get creative!

Pollution Categories

- 1. Plastic and Trash**
- 2. Paper**
- 3. Soil and Yard Waste**
- 4. Special Pollution**

Teacher Preparation

1. Place blue paper, blue plastic, or a shallow layer of water in each tray to create the model stream
2. Add toy fish, plants, rocks, and twigs to create a stream habitat
3. Place simulated pollution throughout the stream and along streambanks
4. Prepare tongs, spoons, or nets for students to use at stations/tables
5. Prepare one model stream for each team

Instructions

1. Show students model streams before they begin
2. Ask students to identify the animals and plants in the habitat
3. Introduce the different cleanup tools and have students take turns using tools to remove one item at a time
4. After removing an item, students identify what type of pollution it represents and place the item in the respective sorting container
5. Students should remove only the pollution and leave fish, plants, rocks and other naturally occurring things
6. After streams have been cleared, briefly discuss how each type of pollution may affect stream habitats and what each student found

Student Observation Sheet

Student Name: _____

Team Number: _____

Before Stream Rescue:

Circle the items you think do not belong in the stream:

- ☐ Paper
- ☐ Plastic
- ☐ Fish
- ☐ Rocks
- ☐ String
- ☐ Plants
- ☐ Soil
- ☐ Oil

Pollution We Found

Pollution Type	Did We Find It?
Paper	Yes/No
Plastic	Yes/No
String	Yes/No
Soil	Yes/No
Oil	Yes/No

One thing I removed was:

One way I can help keep streams clean is:
